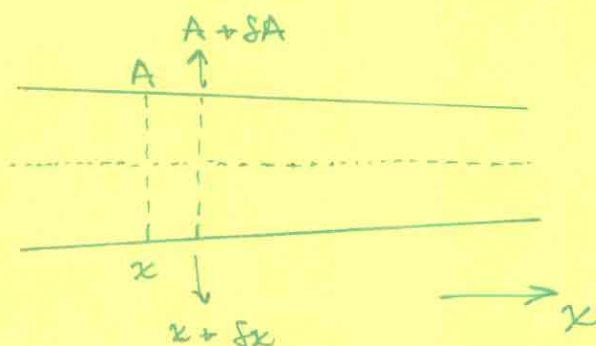


03/12/08

①

Shock Propagation in a Non-uniform Pipe



$$(A \rho u)_{x+\delta x} - (A \rho u)_x$$

$$= - \partial_t (\rho \delta x A)$$

$$\Rightarrow A \delta x \partial_t \rho = - \delta x \partial_x (A \rho u)$$

$$\Rightarrow \partial_t \rho + \partial_x (\rho u) = - \rho u \partial_x \ln A$$

When $\partial_x A = 0$ we recover the usual 1D continuity equation.

We also have the usual dynamic and entropy equations:

$$\frac{du}{dt} = \partial_t u + u \partial_x u = - \frac{1}{\rho} \partial_x p ;$$

$$\frac{dp}{dt} = \left. \frac{\partial p}{\partial s} \right|_{\rho} \frac{ds}{dt} + \left. \frac{\partial p}{\partial \rho} \right|_s \frac{d\rho}{dt} ;$$

$$\text{adiabatic flow} \Rightarrow \frac{ds}{dt} = 0 \Rightarrow \frac{dp}{dt} = c^2 \frac{d\rho}{dt} .$$

So, we have three equations :

$$\partial_t \rho + \partial_x (\rho u) + \rho u \partial_x \ln A = 0 \quad (a)$$

$$\partial_t u + u \partial_x u + \rho^{-1} \partial_x p = 0 \quad (b)$$

$$\partial_t p + u \partial_x p - c^2 (\partial_t \rho + u \partial_x \rho) = 0 \quad (c)$$

(2)

$$(a) \equiv \frac{d\rho}{dt} + \rho \partial_x u + \rho u \partial_x \ln A = 0$$

$$(b) \rightarrow (a) \Rightarrow \frac{1}{c^2} \frac{d\phi}{dt} + \rho \partial_x u + \rho u \partial_x \ln A = 0$$

$$\Rightarrow \partial_t \phi + u \partial_x \phi + \rho c^2 \partial_x u + \rho c^2 u \partial_x \ln A = 0 \quad (cd)$$

$$\rho c \cdot (b) + (cd) =$$

$$\left[\partial_t \phi + (u+c) \partial_x \phi + \rho c [\partial_t u + (u+c) \partial_x u] + \rho c^2 u \partial_x \ln A = 0 \right]$$

$$- \rho c \cdot (b) + (cd) =$$

$$\left[\partial_t \phi + (u-c) \partial_x \phi - \rho c [\partial_t u + (u-c) \partial_x u] + \rho c^2 u \partial_x \ln A = 0 \right]$$

Characteristics $C_{\pm}$ are mapped by trajectories $\frac{dx}{dt} = u \pm c$. Let $(d/dt)_{\pm}$ be the directional time derivatives along $C_{\pm}$:

$$\left(\frac{d}{dt}\right)_{\pm} = \partial_t + (u \pm c) \partial_x$$

$$\Rightarrow \left(\frac{d\phi}{dt}\right)_{+} + \rho c \left(\frac{du}{dt}\right)_{+} + \rho c^2 u \partial_x \ln A = 0$$

$$\left(\frac{d\phi}{dt}\right)_{-} - \rho c \left(\frac{du}{dt}\right)_{-} + \rho c^2 u \partial_x \ln A = 0.$$

When $\partial_x A = 0$ we obtain the usual differential relations

$$d_+ \phi + \rho c d_+ u = 0 \quad (C_+)$$

$$d_- \phi - \rho c d_- u = 0 \quad (C_-)$$

Note that

$$\left(\frac{dA}{dt}\right)_{\pm} = \cancel{\partial_t A} + (u \pm c) \partial_x A = (u \pm c) \partial_x A$$

$$\Rightarrow \partial_x A = \frac{1}{u \pm c} \left(\frac{dA}{dt}\right)_{\pm}$$

Thus, we have

$$\left[\left(\frac{d\phi}{dt}\right)_{\pm} \pm \rho c \left(\frac{du}{dt}\right)_{\pm} + \frac{\rho c^2 u}{u \pm c} \left(\frac{dA}{dt}\right)_{\pm} \frac{1}{A} = 0 \quad (C_{\pm}) \right]$$

These equations give a complete description of the adiabatic ($ds=0$) flow.

Now suppose that the flow we are considering is behind a shock. The shock jump conditions can be written as

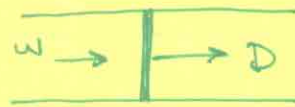
$$\rho_1 u_1 = \rho_0 u_0 = j \quad (\text{Mass})$$

$$p_1 + \rho_1 u_1^2 = p_0 + \rho_0 u_0^2 \quad (\text{Momentum})$$

$$e_1 + \frac{p_1}{\rho_1} + \frac{1}{2} u_1^2 = e_0 + \frac{p_0}{\rho_0} + \frac{1}{2} u_0^2 \quad (\text{Energy})$$

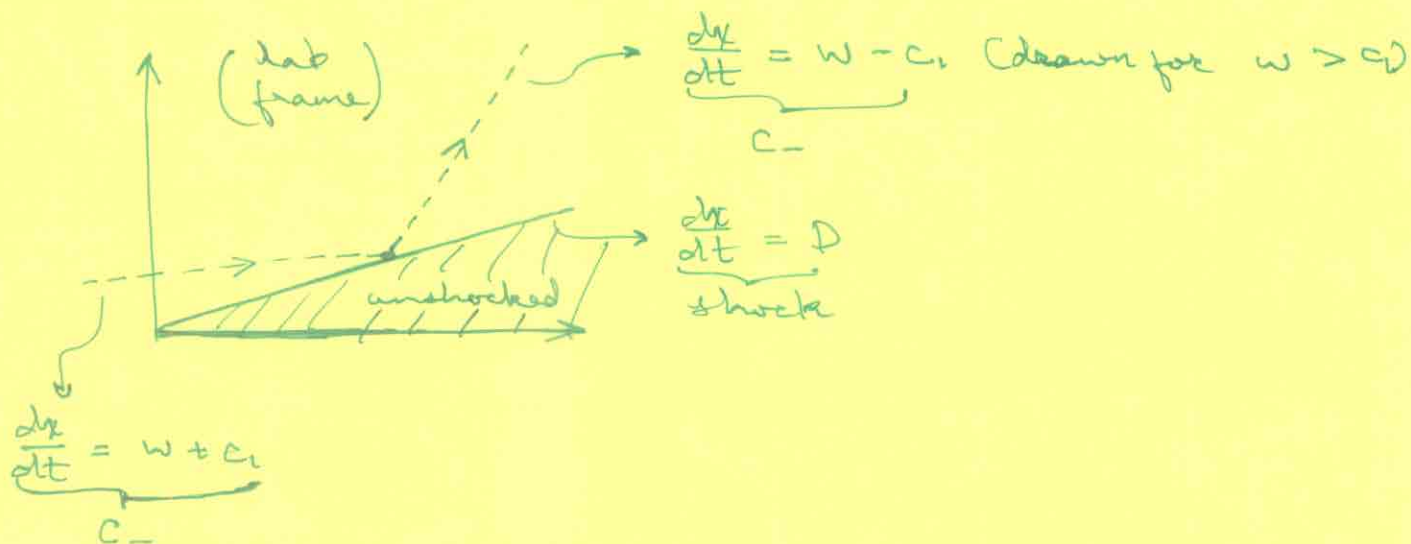


or

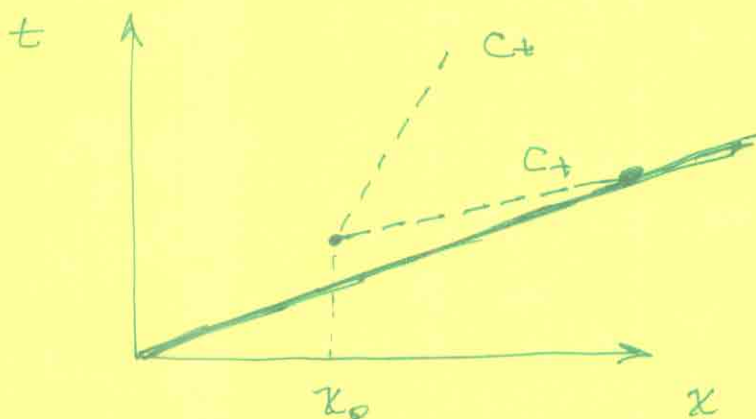
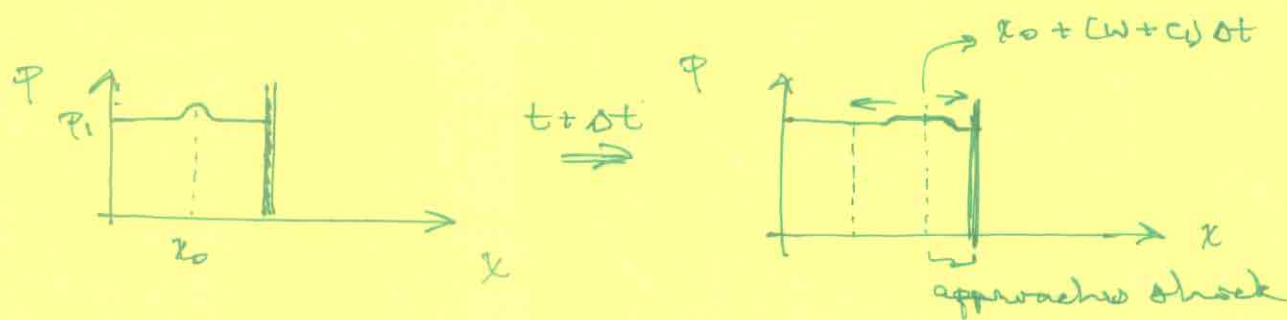


$$\begin{aligned} w_0 &= D \\ w &= w_0 - u_1 \end{aligned}$$

Recall that $u/c_1 < 1 \Rightarrow D - w < c_1$. Therefore, a disturbance moving along C_+ will meet the shock.



A perturbation to the post shock flow will send a disturbance forward and backward (C_+ and C_-), but only the C_+ meets the shock and can have a causal influence.



Along C_+ , changes in p , u , and A are related by

$$dp + \rho c du + \frac{\rho c^2 u}{u + c} d \ln A = 0.$$

A perturbation to the post shock flow is carried to the shock by the above equation. The post-shock fluid variables are modified as a result, and the shock speed D must change as a consequence. In particular, a positive pressure perturbation, with $\delta p = \delta p / c_i^2 > 0$, will cause the post-shock conditions to move up the H curve, giving a steeper slope $j^2 = \frac{D^2}{V_0^2} = -\frac{\Delta p}{\Delta V}$.

In this interpretation, we could imagine substituting for dp and du in the C_+ equation the corresponding variations in the shock jump conditions:

$$\text{let } dp \rightarrow dp_1, \quad du \rightarrow dw; \quad \underbrace{\rho \rightarrow \rho_1, c \rightarrow c_1}_{\text{unperturbed background}}; \\ u \rightarrow w$$

$$\Rightarrow dp_1 + \rho_1 c_1 dw + \frac{\rho_1 c_1^2 w}{w + c_1} d \ln A = 0.$$

The hypothesis is that this equation describes how the post-shock conditions vary with small changes in A .

For the shock jump, we have the variations

$$\rho_1 u_1 + p_1 du_1 = p_0 du_0$$

$$\begin{aligned} dp_1 + \rho_1 u_1^2 + 2\rho_1 u_1 du_1 &= 2p_0 du_0 \\ \Rightarrow dp_1 + \rho_1 u_1^2 &= 2j \underbrace{(du_0 - du_1)}_{= du} \end{aligned}$$

In deriving these equations we have $p_0 = \text{const}$ and $p_0 = \text{const}$ upstream; since $\Phi > c_0$, the upstream flow is unaffected by the shock until it passes.

Note that $d\phi_1$ and $d\rho_1$ are related by their mutual evolution along the H curve. Following Landau & Lifshitz, let

$$h = - \left(\frac{j}{\rho_1} \right)^2 \frac{d\rho}{d\phi} \Big|_H = j^2 \frac{dN}{d\phi} \Big|_H$$

Now,

$$d\rho_1 = -h \left(\frac{\rho_1}{j} \right)^2 d\phi_1$$

$$\Rightarrow -h \frac{\rho_1^2}{j^2} d\phi_1 u_1 + \rho_1 du_1 = p_0 du_0$$

$$d\phi_1 + u_1^2 \left(-h \frac{\rho_1^2}{j^2} \right) d\phi_1 = 2j (du_0 - du_1)$$

$$\Rightarrow du_1 = \frac{1}{e_1} [e_0 du_0 + h \frac{e_1^2}{j^2} u_1 d\varphi_1]$$

$$\begin{aligned} \Rightarrow d\varphi_1 \left[1 - \cancel{u_1^2} h \frac{e_1^2}{j^2} \right] &= 2j du_0 - 2j \frac{1}{e_1} [e_0 du_0 + h \frac{e_1^2}{j^2} u_1 d\varphi_1] \\ &= 2j du_0 - \frac{2j e_0}{e_1} du_0 - 2h \frac{e_1}{j} u_1 d\varphi_1 \end{aligned}$$

$$\Rightarrow d\varphi_1 [1 - h + 2h] = 2j \left(1 - \frac{e_0}{e_1}\right) du_0$$

$$\Rightarrow \boxed{d\varphi_1 = \frac{1}{1+h} \cdot 2j \left(1 - \frac{e_0}{e_1}\right) du_0}$$

$$\begin{aligned} \Rightarrow du_1 &= \frac{e_0}{e_1} du_0 + h \frac{1}{j} \left\{ \frac{2j}{1+h} \left(1 - \frac{e_0}{e_1}\right) du_0 \right\} \\ &= \boxed{du_0 \left\{ \frac{e_0}{e_1} + \frac{2h}{1+h} \left(1 - \frac{e_0}{e_1}\right) \right\}} \end{aligned}$$

$$\begin{aligned} du_0 - du_1 &= du_0 \left\{ 1 - \frac{e_0}{e_1} - \frac{2h}{1+h} \left(1 - \frac{e_0}{e_1}\right) \right\} \\ &= du_0 \left(1 - \frac{e_0}{e_1}\right) \left(1 - \frac{2h}{1+h}\right) \\ &= du_0 \left(1 - \frac{e_0}{e_1}\right) \left(\frac{1-h}{1+h}\right) \end{aligned}$$

$$\Rightarrow \text{(from C+): } d\varphi_1 + e_1 c_1 (du_0 - du_1) =$$

$$\frac{2j}{1+h} \left(1 - \frac{e_0}{e_1}\right) du_0 + e_1 c_1 du_0 \left(1 - \frac{e_0}{e_1}\right) \left(\frac{1-h}{1+h}\right)$$

$$= du_0 \left(1 - \frac{e_0}{e_1}\right) \frac{1}{1+h} \left\{ \cancel{2j}^{e_1 u_1} + e_1 c_1 (1-h) \right\}$$

$$= du_0 \left(1 - \frac{e_0}{e_1}\right) \frac{e_1 c_1}{1+h} \left\{ 2^{u_1/c_1} + 1 - h \right\}$$

Now the C_+ equation becomes

$$du_0 \left(1 - \frac{p_0}{e_1}\right) \frac{e_1 e_1}{1+h} \left[2 \frac{u_1}{c_1} + 1 - h\right] + \frac{e_1 c_1^2 (u_0 - u_1)}{u_0 - u_1 + c_1} d \ln A = 0$$

$$\Rightarrow - \frac{d \ln A}{du_0} = \frac{u_0 - u_1 + c_1}{c_1 (u_0 - u_1)} \left(1 - \frac{p_0}{e_1}\right) \left(\frac{2 u_1}{c_1} + 1 - h\right)$$

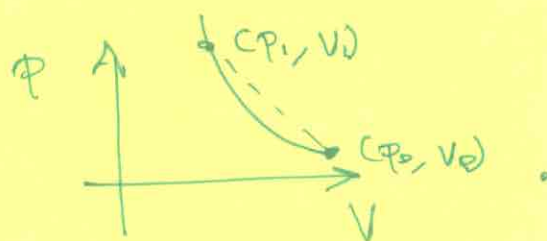
Note: $\frac{1}{u_0 - u_1} \left(1 - \frac{p_0}{e_1}\right) = \frac{e_1 - p_0}{e_1 u_0 - e_1 u_1} \cdot \frac{u_0}{u_0}$

$$= \frac{(u_0 e_1 - p_0 u_0)}{(e_1 u_0 - e_1 u_1)} \frac{1}{u_0} = \frac{1}{u_0}$$

$$\Rightarrow - \frac{d \ln A}{du_0} = \underbrace{\frac{(u_0 - u_1) + c_1}{c_1 u_0}}_{> 0} \left(\frac{2 u_1}{c_1} + 1 - h\right)$$

In most cases of interest,

$$h = j^2 \frac{dW}{dp} \Big|_H < 0$$



For a strong shock, $(P_1/P_0 \gg 1)$ ($j = \text{const}$)

$$\frac{u_1}{V_0} \rightarrow \frac{1-j}{1+j} \Rightarrow \frac{dW}{dp_1} = 0$$

$$\Rightarrow \text{sign} \left[- \frac{d \ln A}{du_0} \right] = \text{sign} \left[\frac{2 u_1}{c_1} + 1 \right] \geq 0$$

$\Rightarrow$ If A decreases, u_0 increases.

For a weak shock ($p_1/p_0 - 1 \ll 1$) with $\gamma = \text{const}$:

$$h = \gamma^2 \frac{dV_1}{dp_1} = - \frac{\gamma p}{\gamma V} \frac{dV_1}{dp_1} \rightarrow \underline{-1} + O(\epsilon)$$

It can also be shown that

$$u_1^2 = c_1^2 \frac{1}{\gamma} [\gamma - 1 + (\gamma + 1) \frac{p_0}{p_1}]$$

$$\frac{p_1}{p_0} = 1 + \epsilon \Rightarrow u_1^2 \approx c_1^2 \frac{1}{\gamma} [\gamma - 1 + (\gamma + 1)(1 - \epsilon)]$$

$$= c_1^2 \frac{1}{\gamma} [\gamma - 1 + 2\gamma - (\gamma + 1)\epsilon] = c_1^2 [1 - \frac{\gamma + 1}{\gamma} \epsilon]$$

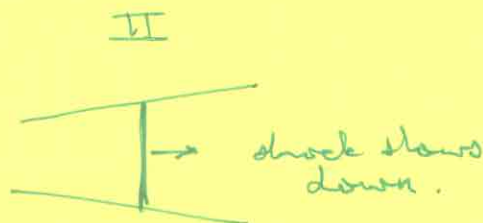
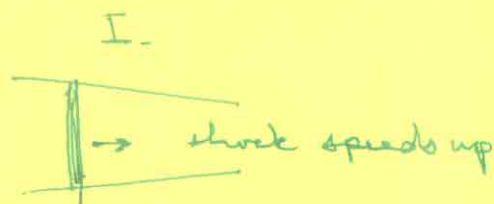
$$\Rightarrow \frac{u_1^2}{c_1^2} \approx 1 - \frac{\gamma + 1}{\gamma} \epsilon > 0$$

$$\Rightarrow \frac{2 \frac{u_1}{c_1} + 1 - h}{1 + h} \rightarrow \frac{2 - \frac{\gamma + 1}{\gamma} \epsilon + 1 + 1}{1 + 1} (1 + h)$$

$1 + h = 1 - |h|$; since $|h| \in (-1, 0)$ from (weak, strong)

$$\Rightarrow - \frac{d \ln A}{d \ln p} > 0 \quad \text{for weak shocks as well.}$$

It can be shown, in general, that an ideal gas with $\gamma = \text{const}$ always has $d \ln A / d p < 0$.

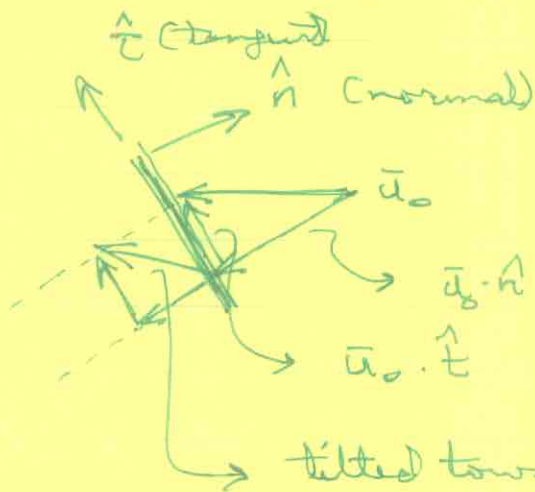


What does this have to do with shock curvature?

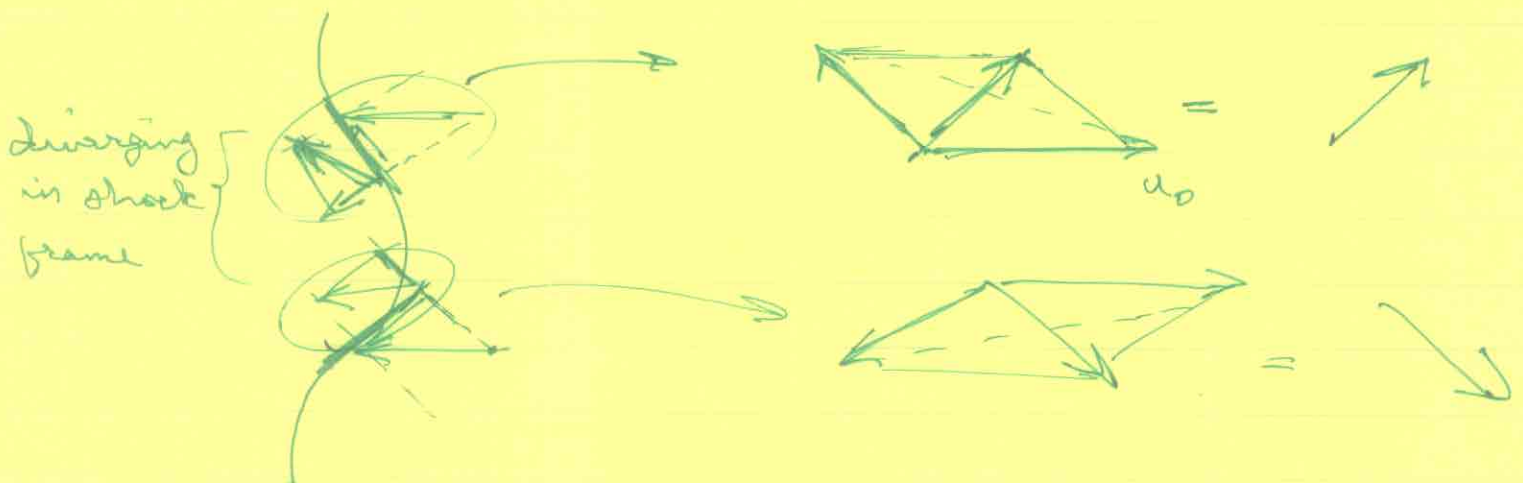
Consider a small perturbation to the shape of a planar shock:



In a small patch of the perturbed front:



$\bar{u}_0 \cdot \hat{t}$ preserved
 $\bar{u}_0 \cdot \hat{n}$ not preserved
 (accelerated)



$\Rightarrow$ diverging flow $\Rightarrow \underline{d \ln A} > 0$